
Reducing Inter-Class Confusion in Military Aircraft Detection via Selective Oversampling

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Abstract

Military aircraft detection in long-range aerial reconnaissance imagery often suffers from inter-class confusion among visually similar categories, notably between F-series fighters and European delta-wing aircraft. Although generative AI-based augmentation has received increasing attention, the impact of rebalancing the original dataset prior to generating synthetic samples remains underexplored. In this work, we investigate selective oversampling targeted at predefined confusion pairs. We first examine the relationship between object size and detection performance, and the results indicate that object size is not the primary factor behind the observed detection failures. We then apply a confusion-aware rebalancing strategy to three representative confusion pairs (F15↔F14, F16↔F18, and Eurofighter Typhoon (EF2000)↔Rafale). Confusion-aware rebalancing reweights the training data by selectively oversampling images that contain the target confusion pairs. The sampling ratio is set using pairwise confusion rates measured from a baseline model. Experimental results show that selective oversampling can improve mAP@0.5 and reduce misclassification for some pairs, but may worsen pairwise confusion when the visual similarity between classes is extremely high. These findings suggest that selective oversampling is an important component of data rebalancing, but is insufficient on its own to fully resolve inter-class confusion in military aircraft detection.

1 Introduction

1.1 Background and Motivation

Detecting military aircraft in long-range aerial images is challenging. This is mainly because targets are small and visually similar. For example, some F-series fighters (e.g., F-15 vs. F-14) and European fighter aircraft (e.g., EF2000 vs. Rafale) exhibit very similar silhouettes. Such similarity increases inter-class confusion and leads to unstable detection performance. In operational settings, misidentifying a target can directly lead to severe decision-making errors.

1.2 Limitations of Existing Augmentation Approaches

To address the lack of training data, researchers use generative AI (e.g., Stable Diffusion) to create synthetic images. However, synthetic data may offer limited benefits if underlying issues in the real dataset (e.g., class imbalance or label noise) remain unresolved. It remains unclear whether failures stem from insufficient data or from extreme visual similarity between aircraft. Therefore, before adopting generative augmentation, we first evaluate whether confusion can be reduced using the existing data alone.

1.3 Contributions and Study Scope

The main contributions of this study are as follows.

1. **Size vs. Shape:** To address the lack of training data, researchers use generative AI (e.g., Stable Diffusion) to create synthetic images. However, synthetic data may offer limited benefits if underlying issues in the real dataset (e.g., class imbalance or label noise) remain unresolved. It remains unclear whether failures stem from insufficient data or from extreme visual similarity between aircraft. Therefore, before adopting generative augmentation, we first evaluate whether confusion can be reduced using the existing data alone.
2. **Testing Selective Oversampling:** We tested “selective oversampling” targeted at predefined confusion pairs. This helps moderately similar pairs (F-15 vs. F-14) but can increase pairwise misclassification for extremely similar pairs (EF2000 vs. Rafale).
3. **Metric Analysis:** We show that a higher mAP (e.g., mAP@0.5) does not necessarily imply lower inter-class confusion. This suggests that standard metrics can obscure pairwise errors in fine-grained classification.

1.4 Contribution Clarification

We analyze when data rebalancing succeeds and when it fails. Results suggest that simple oversampling is insufficient for extremely similar targets, indicating that more advanced approaches are needed in future work.

2 Related works

2.1 Small Object Detection in Aerial Imagery

Detecting small objects in aerial images is notoriously difficult compared to natural scene imagery due to the loss of high-frequency details during down-sampling. Nikouei et al. [2] explain that this requires specialized feature extraction methods beyond standard CNNs. To address this, Kim et al. [1] proposed enhanced strategies, specifically input resolution adjustment and attention mechanisms, to improve the detection of “tiny” targets. Furthermore, within the defense domain, Mukherjee et al. [3] utilized the state-of-the-art YOLOv10 architecture to classify military assets, emphasizing that the core challenge is not just localization but distinguishing between visually similar airframes (e.g., Rafale vs. F-35) in operational environments.

Recent surveys further confirm that small object detection in aerial and remote sensing imagery remains fundamentally constrained by limited spatial resolution and weak discriminative features. Wang et al. [4] systematically reviewed deep learning-based small object detection methods for remote sensing and reported that most performance gains originate from scale-aware feature enhancement rather than improved class discrimination. Similarly, Hua and Chen [5] emphasized that despite advances in multi-scale architectures and attention mechanisms, visually similar objects often remain indistinguishable due to insufficient semantic separation at the feature level. These studies suggest that current approaches predominantly focus on improving detectability, while inter-class confusion caused by visual similarity is largely left unaddressed.

2.2 Class Imbalance and Resampling Strategies

For real-time detection, the YOLO framework is the de facto standard. Ramos and Sappa [6] reviewed a decade of YOLO's evolution, noting that while speed has improved, performance often degrades on “long-tailed” datasets where minority classes are under-represented. In such cases, standard models tend to bias towards majority classes. To mitigate this, Shorten and Khoshgoftaar [7] argued that data augmentation is essential. However, generic augmentation techniques often fail to resolve the specific decision boundary instability between fine-grained confusion pairs, necessitating more targeted rebalancing strategies.

Beyond data augmentation, several studies have addressed class imbalance through loss reweighting rather than resampling. Lin et al. [8] introduced Focal Loss to reduce the dominance of easy majority samples during training, while Cui et al. [9] proposed class-balanced loss based on the effective

number of samples to mitigate long-tailed distributions. However, these approaches operate at the class level and do not explicitly account for pairwise inter-class confusion or decision boundary instability between visually similar categories.

2.3 Generative Data Augmentation for Aircraft Detection

Acquiring labeled data for rare military targets is limited by security and cost. To overcome this, researchers are utilizing synthetic data generation. Kim et al. [10] proposed a methodology using the military simulation game *Arma 3* to construct large-scale training datasets. While they demonstrated that synthetic data can supplement real-world imagery, relying on it introduces a "domain gap" between synthetic rendering and real-world physics. Therefore, our study investigates whether selective oversampling can effectively utilize the existing real data distribution to resolve confusion, serving as a prerequisite step before introducing synthetic artifacts.

Prior studies have shown that synthetic data can effectively supplement limited real-world samples for object detection. Tremblay et al. [11] demonstrated that large-scale synthetic data improves detector performance, but also highlighted the persistent reality gap caused by differences in texture, illumination, and sensor characteristics. Similarly, Peng et al. [12] showed that detectors trained on 3D model-based synthetic data require real images to achieve robust generalization. These findings indicate that while generative augmentation is promising, understanding and stabilizing the real data distribution remains a necessary prerequisite before introducing synthetic samples.

3 Methodology and Experimental Setup

3.1 Dataset and Class Distribution

In this study, we utilized an aerial reconnaissance dataset containing multiple classes of military aircraft. The entire dataset was partitioned into training and validation sets with a 9:1 ratio.

Table 1: Class-wise Image Distribution in the Training and Validation Sets

	Class Name	Train Images	Validation Images	Total
Fighter (US)	F-14	469	78	547
	F-15	1,435	106	1,541
	F-16	1,686	255	1,941
	F-18	1,630	120	1,750
Fighter (EU)	EF2000	746	88	834
	Rafale	741	59	800
Others	Other Classes (N=90)	28,813	3,256	32,069
-	Total	35,520	3,962	39,482

As summarized in Table 1, the dataset exhibits significant class imbalance. While certain fighter classes (e.g., F-16, F-18) are represented by a relatively large number of samples, others suffer from limited data availability. Specifically, the targeted confusion pairs (F15↔F14, F16↔F18, and EF2000↔Rafale) show distinct disparities not only in sample counts but also in data distribution characteristics. This imbalance contributes to noticeable variance in class-wise mAP@0.5 performance within the baseline model. These observations suggest that performance gaps are primarily attributed to data distribution properties rather than the limitations of the model architecture itself, motivating the adoption of a confusion-pair-focused selective oversampling strategy rather than uniform augmentation.

3.2 Experimental Environment and Implementation Details

To ensure a fair comparison between the baseline and the selective oversampling models, all hardware and software conditions were kept identical. Experiments were conducted on a workstation equipped with a single NVIDIA GeForce RTX 3060 GPU. We employed YOLOv11m[13] as the

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baseline one-stage object detector. The training configuration was standardized as follows:

Table 2: Implementation Details and Hyperparameters

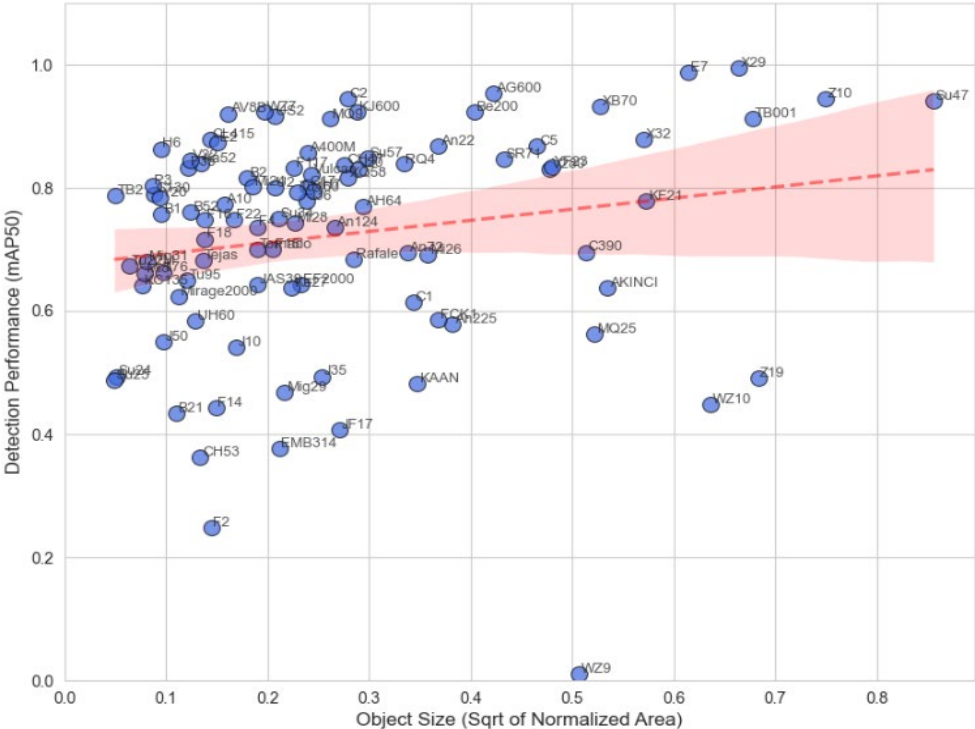
Category	Parameter	Value
Hardware	GPU	NVIDIA GeForce RTX 3060 (Single)
Model	Architecture	YOLOv11m (One-stage detector)
	Input Resolution	640 x 640
	Pre-trained Weights	COCO Pre-trained
Training	Batch Size	16
	Total Epochs	50
	Optimizer	SGD (Default settings)
	Workers	2
	Device	0

To isolate the effect of the rebalancing strategy, standard data augmentation techniques (e.g., Mosaic, Mixup) inherent to the YOLO framework were maintained across all runs. Selective oversampling was applied strictly during the construction of the training dataset, targeting only the identified confusion pairs. The validation set remained in its original distribution to reflect real-world operational conditions. Performance was evaluated using class-wise $mAP@0.5$ and the aggregate count of bidirectional misclassifications within the confusion pairs.

4 Impact of Object Scale on Detection Performance

To investigate the effect of scale, we defined object size as the square root of the normalized bounding box area. Figure 1 illustrates the correlation between this metric and detection performance ($mAP@0.5$) across all classes.

Figure 1: Impact of Object Size on Detection performance (Correlation : 0.18)



As shown in Figure 1, the correlation between object size and performance is weak (0.18). Although a slight trend exists where larger objects are detected better, numerous small objects still exhibit high mAP, whereas certain larger objects (e.g., specific transport aircraft) suffer from low performance. This discrepancy indicates that morphological similarity, rather than object scale, is the dominant factor driving detection failures in this dataset. These findings motivate a shift from scale-focused remedies toward confusion-aware data rebalancing strategies that explicitly target visually ambiguous class pairs.

5 Confusion Pair Analysis

5.1 Definition of Confusion Pairs

To identify class pairs that suffer from severe inter-class confusion, we analyzed the misclassification statistics of the baseline model on the validation set. Figure 2 presents the Top-10 confusion pairs ranked by the total number of bidirectional misclassifications.

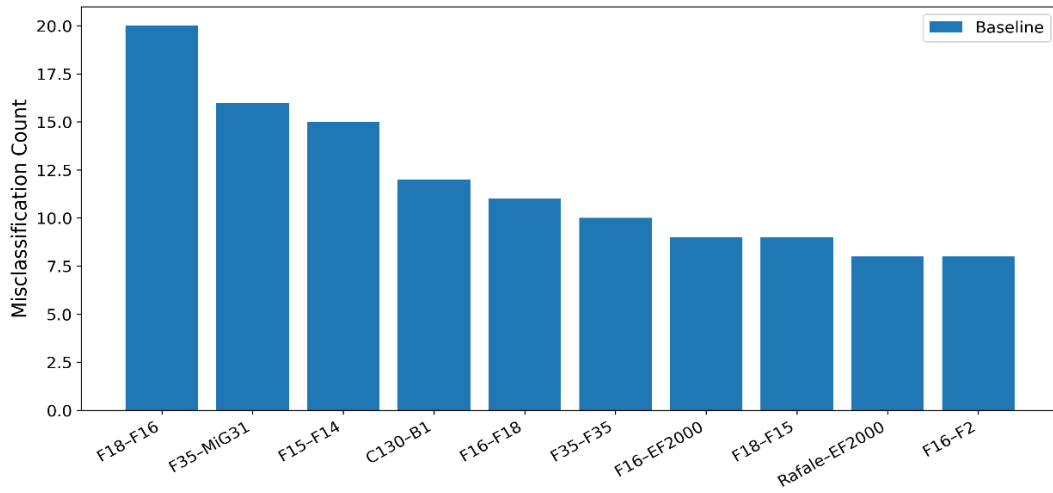


Figure 2: Top-10 Confusion Pairs (Baseline)

This figure shows the most frequent bidirectional misclassification pairs on the validation set. Each bar represents the total number of misclassifications in both directions ($A \rightarrow B$ and $B \rightarrow A$), indicating class pairs with severe inter-class confusion before any rebalancing.

Based on this analysis, we selected representative confusion pairs by considering both misclassification frequency and visual similarity between aircraft types. Among the candidates, three pairs—F15 $\leftrightarrow$ F14, F16 $\leftrightarrow$ F18, and EF2000 $\leftrightarrow$ Rafale—were chosen for detailed analysis because they consistently showed high confusion and shared similar silhouettes and structural features.

Based on the validation results of the baseline model, we define confusion pairs as class pairs with frequent mutual misclassifications. From the top five candidates, three pairs were finally selected by considering both misclassification frequency and visual similarity.

5.2 Representative Confusion Pairs

The selected representative confusion pairs are as follows:

· F15 $\leftrightarrow$ F14 / · F16 $\leftrightarrow$ F18 / · EF2000 $\leftrightarrow$ Rafale

5.3 Visual Examples of Confusion Pairs

For each confusion pair, we present real misclassification examples to visually illustrate the similarity between classes.



Figure 3: Representative Visual Examples for Inter-Class Confusion Pairs

Figure 3 shows representative misclassification examples from the baseline model for the three selected confusion pairs. Each image is annotated with the ground-truth class, the predicted class, and the corresponding confidence score.

The $F15 \leftrightarrow F14$ and $F16 \leftrightarrow F18$ pairs exhibit confusion under specific viewpoints or partial occlusions, while still preserving some discriminative visual cues. In contrast, the $EF2000 \leftrightarrow Rafale$ pair demonstrates severe visual ambiguity, where even high-confidence predictions fail to distinguish between the two classes due to their nearly identical silhouettes and structural characteristics.

6 Confusion-Aware Selective Oversampling Method

6.1 Motivation

Uniformly oversampling all classes can cause unnecessary data duplication and reduce training efficiency, particularly when misclassification errors are concentrated in a limited number of visually similar class pairs. In such cases, global rebalancing may introduce redundant samples for well-learned classes while failing to directly address unstable decision boundaries formed between highly confused classes. This observation motivates a more targeted rebalancing strategy that focuses explicitly on empirically identified confusion regions.

6.2 Oversampling Strategy

Selective oversampling was applied exclusively to classes involved in the identified confusion pairs. Rather than enforcing global class balance, the strategy aims to reduce representational disparity within each confusion pair by selectively increasing the number of training samples for underrepresented classes.

Table 3: Selective Oversampling Ratios for Identified Confusion Classes

Class	Train Images	Validation Images	Oversampling Ratio	Effective Train Images
F-14	469	78	$\times 2.5$	$\approx 1,170$
F-15	1,435	106	$\times 1.0$	1,435
F-16	1,686	255	$\times 1.0$	1,686
F-18	1,630	120	$\times 1.0$	1,630
EF2000	746	88	$\times 1.5$	$\approx 1,120$
Rafale	741	59	$\times 1.5$	$\approx 1,110$

Note: Oversampling was applied only to the training set, the validation set was kept unchanged to preserve the original data distribution.

The oversampling ratios were determined to approximately align the effective number of training samples within each confusion pair, while leaving sufficiently represented classes unchanged. This design avoids unnecessary duplication of well-learned classes and focuses the rebalancing effort on regions where decision boundaries were empirically observed to be unstable.

6.3 Expected Effect

By selectively increasing training samples only for underrepresented classes within confusion pairs, the proposed method is expected to stabilize local decision boundaries without introducing excessive redundancy across the entire dataset. This targeted rebalancing operates at the data distribution level and complements the fixed model architecture, enabling performance gains without additional model complexity.

7 Experimental Results

7.1 Evaluation Metrics

Performance evaluation was conducted using class-wise [mAP@0.5](#) and the number of bidirectional misclassifications within predefined confusion pairs.

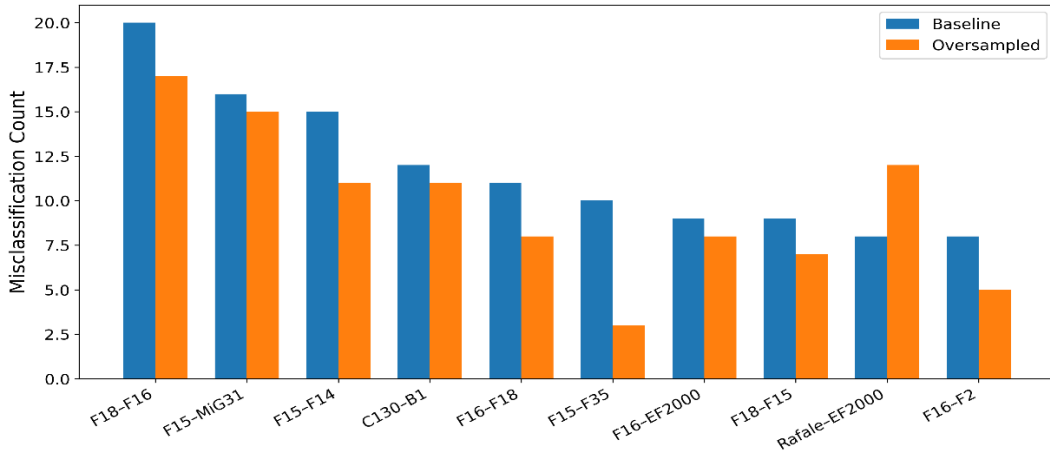


Figure 4: Comparison of Top-10 Confusion Pairs Between Baseline and Oversampled Models

The figure reports the total number of bidirectional misclassifications for each confusion pair, evaluated on the same validation set. While selective oversampling reduces confusion for several pairs, the effect is not uniform across all classes, and some pairs exhibit increased confusion after rebalancing.

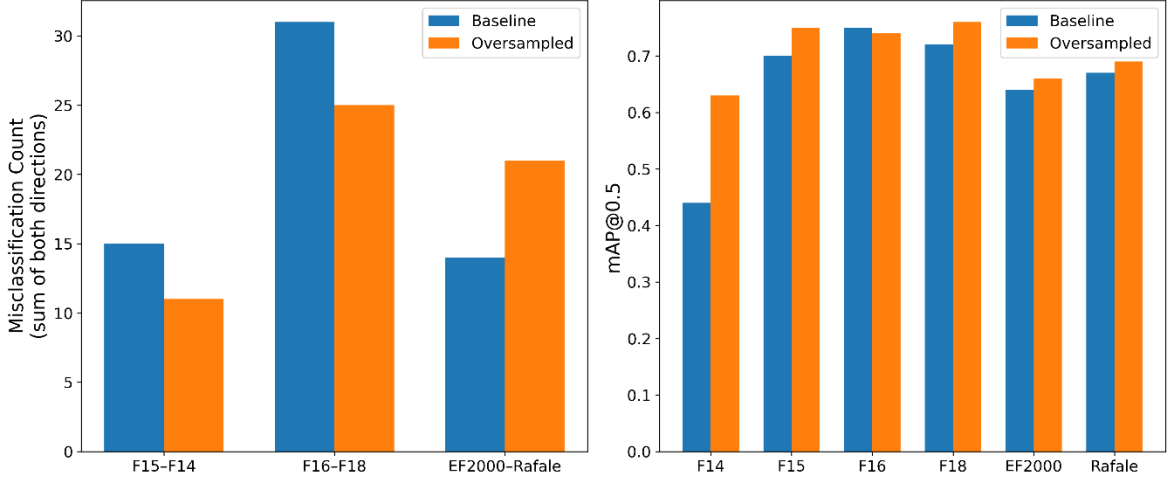


Figure 5: Effect of selective oversampling on inter-class confusion

The left panel shows aggregated bidirectional misclassification counts for three representative confusion pairs before and after selective oversampling. The right panel presents the corresponding class-wise [mAP@0.5](#) for the six related aircraft classes. Selective oversampling reduces confusion for some pairs, but its effect is not consistent across all cases.

Figure 5 illustrates how selective oversampling affects both inter-class confusion and class-wise detection performance for the selected confusion pairs. For the F15↔F14 pair, selective oversampling reduces the total number of bidirectional misclassifications from 15 to 11, along with clear improvements in mAP@0.5 for both classes. A similar trend is observed for the F16↔F18 pair, where confusion decreases from 31 to 25.

In contrast, the EF2000↔Rafale pair shows the opposite behavior. Although mAP@0.5 slightly increases for both EF2000 and Rafale, the total number of bidirectional misclassifications increases from 14 to 21 after oversampling. This result indicates that improvements in aggregate detection metrics do not necessarily lead to reduced inter-class confusion, especially for aircraft pairs with extreme morphological similarity.

8 Discussion

The experimental results clearly show that selective oversampling does not consistently reduce inter-class confusion across all classes. For class pairs with a moderate level of visual similarity, such as F15↔F14 and F16↔F18, selective oversampling increases data density and strengthens existing discriminative visual features. As a result, the decision boundaries become more stable. This suggests that confusion in these pairs was mainly caused by data imbalance or insufficient exposure during training.

In contrast, different behavior is observed for class pairs with extremely high morphological similarity, such as EF2000↔Rafale. In this case, repeated exposure through oversampling does not introduce new discriminative cues. Instead, it reinforces already ambiguous visual patterns, which leads to increased confusion. This reflects a structural limitation of oversampling: while it can rebalance data distributions, it cannot create discriminative features that do not exist in the original data.

This study also shows that aggregate performance metrics such as mAP@0.5 may not fully explain inter-class confusion. In scenarios involving visually similar targets, improvements in class-wise detection performance do not necessarily lead to more stable decision boundaries or fewer misclassifications. This observation highlights the need for confusion-aware evaluation to complement evaluations based on single performance metrics.

From this perspective, selective oversampling can be considered a necessary condition for reducing confusion, but it is not a sufficient condition. Future work should focus on more fundamental solutions, such as guided generative synthesis or representation-level augmentation, which can explicitly control morphological differences while preserving object size and overall data distribution. These approaches may better address confusion caused by extreme visual similarity.

9 Conclusion

This study systematically analyzes the effectiveness and limitations of confusion-aware selective oversampling for military aircraft detection in long-range aerial images. By first examining the relationship between object size and detection performance, we confirm that object size is not the primary cause of detection failure in this dataset. Based on this finding, selective oversampling is applied to representative confusion pairs, and both changes in inter-class confusion and class-wise detection performance are analyzed.

The experimental results show that for class pairs with a moderate level of visual similarity, selective oversampling improves class-wise mAP@0.5 while also reducing inter-class confusion. In contrast, for class pairs with extreme morphological similarity, such as EF2000↔Rafale, confusion increases despite marginal improvements in mAP@0.5. These results clearly demonstrate that selective oversampling may be a necessary condition for confusion reduction, but it is not sufficient to resolve structural ambiguity between highly similar classes.

Overall, this study provides experimental evidence that data rebalancing alone cannot fundamentally resolve confusion between visually indistinguishable classes. Future research should focus on introducing new discriminative cues through guided generative synthesis or representation-level augmentation methods, which can explicitly control morphological differences while preserving the overall data distribution.

10 References

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1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [YES]

Justification:

The abstract accurately summarizes the mixed results (improvement in mAP vs. degradation in EF2000 / Rafale confusion) found in the experiments.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification:

The Discussion section explicitly discusses the failure of oversampling for the EF2000/Rafale pair due to extreme visual similarity.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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The dataset is publicly available at Kaggle.
(<https://www.kaggle.com/datasets/a2015003713/militaryaircraftdetectiondataset>)

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Section 3 details the model architecture and training hyperparameters.

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The analysis relies on a fixed validation set comparison; multiple runs/error bars are not reported.

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